CHAPTER 1

UNITS AND MEASUREMENTS

CHAPTER ESSENCE AT A GLANCE:
Physics is the language of measurement! Everything in the universe boils down to just 7 fundamental base units (like Mass, Length, Time). Any equation is valid ONLY if both sides have matching dimensional powers. It's like checking that apples equal apples!

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Base vs Derived Units:** 7 Base quantities (kg, m, s, A, K, mol, cd) are defined independently; Derived quantities (Speed, Force, Energy) are combinations of base units.

• **Principle of Homogeneity:** In any valid physical equation, dimensions on LHS MUST equal dimensions on RHS ([LHS] = [RHS]). You can only add or subtract quantities with identical dimensions!

• **Spring Constant k:** Restoring force per unit elongation: F = -k x implies [k] = [F] divided by [x] = [M L T^{-2}] divided by [L] = [M T^{-2}].

• **Einstein Relativistic Missing 'c':** In m = m_0 divided by sqrt(1 - v^2), the subtracted term (1 - v^2) must be dimensionless. Since [v^2] = [L^2 T^{-2}], it must be divided by c^2: m = m_0 divided by sqrt(1 - v^2 divided by c^2).

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$[G] = \frac{[F][r^2]}{[m_1][m_2]} = [M^{-1}L^{3}T^{-2}]$$

Dimensional formula of Gravitational Constant G

$$[h] = \frac{[E]}{[\nu]} = [ML^{2}T^{-1}]$$

Dimensional formula of Planck's Constant h

$$[\eta] = \frac{[F]}{[A]\frac{dv}{dx}} = [ML^{-1}T^{-1}]$$

Dimensional formula of Viscosity Coefficient

$$T = 2\pi\sqrt{\frac{l}{g}}$$

Period of simple pendulum via dimensional analysis

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Never confuse units with dimensions! Angle has a unit (radian) but has NO dimensions (dimensionless). In exam questions, check whether the power of [T] is -1 or -2!

• **Alert:** If you forget any dimensional formula in the exam, write its simplest formula (e.g., Work = Force × Distance = M L T^{-2} × L = M L^2 T^{-2}) in 5 seconds!

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. State the Principle of Homogeneity of dimensions. Illustrate with an example. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Statement]** The Principle of Homogeneity states that the dimensions of each term on both sides of a physically valid equation must be identical. Only physical quantities having identical dimensions can be added, subtracted, or equated. [1 Mark]

• **[Step 2: Example]** In the kinematic equation s = ut + half (0.5) at^2: [s] = [L], [ut] = [L T^{-1}][T] = [L], and [half (0.5) at^2] = [L T^{-2}][T^2] = [L]. Every term has dimension [L]. [1 Mark]

• **Key Exam Result:** [LHS] = [RHS] for every additive term

2. Write the dimensional formula of: (a) Universal Gravitational Constant (G) (b) Planck's constant (h). [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Gravitational Constant G]** From F = G m1 m2 divided by r^2, G = F r^2 divided by (m1 m2). [G] = [M L T^{-2}][L^2] divided by [M^2] = [M^{-1} L^3 T^{-2}]. [1 Mark]

• **[Step 2: Planck's Constant h]** From E = h ν, h = E divided by ν. [h] = [M L^2 T^{-2}] divided by [T^{-1}] = [M L^2 T^{-1}]. [1 Mark]

• **Key Exam Result:** [G] = M^{-1} L^3 T^{-2}, [h] = M L^2 T^{-1}

3. Check the dimensional consistency of the following equations: (a) $v^2 - u^2 = 2as$ (b) $T = 2\pi \sqrt{l/g}$. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Check $v^2 - u^2 = 2as$]** [v^2] = ([L T^{-1}])^2 = [L^2 T^{-2}]. [u^2] = [L^2 T^{-2}]. [2as] = [L T^{-2}][L] = [L^2 T^{-2}]. Since [LHS] = [RHS] = [L^2 T^{-2}], the equation is dimensionally consistent. [1.5 Marks]

• **[Step 2: Check $T = 2\pi\sqrt{l/g}$]** [LHS] = [T] = [T^1]. [RHS] = √([l] divided by [g]) = √([L] divided by [L T^{-2}]) = √([T^2]) = [T^1]. Since [LHS] = [RHS] = [T], the equation is dimensionally consistent. [1.5 Marks]

• **Key Exam Result:** Both equations are dimensionally consistent

4. The centripetal force F acting on a particle depends on its mass m, speed v, and radius r of circular path. Derive an expression for F using dimensional analysis. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Setup Equation]** Let F = k m^a v^b r^c, where k is a dimensionless constant. Dimensional formulas: [F] = [M L T^{-2}], [m] = [M], [v] = [L T^{-1}], [r] = [L]. [M^1 L^1 T^{-2}] = [M]^a [L T^{-1}]^b [L]^c = [M^a L^(b+c) T^-b]. [1 Mark]

• **[Step 2: Equating Powers]** Equating powers of M, L, T on both sides: a = 1; -b = -2 => b = 2; b + c = 1 => 2 + c = 1 => c = -1. [1.5 Marks]

• **[Step 3: Final Expression]** Substituting: F = k m^1 v^2 r^{-1} = k m v^2 divided by r. Experimentally k = 1, therefore F = m v^2 divided by r. [0.5 Mark]

• **Key Exam Result:** Centripetal Force F = m v^2 divided by r

5. Convert an energy of 1 Joule into ergs using the method of dimensional analysis. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Dimensional Formula of Energy]** [Energy] = [M^1 L^2 T^{-2}]. Here a = 1, b = 2, c = -2. [1 Mark]

• **[Step 2: System Comparison]** In SI (Joule): M1 = 1 kg = 1000 g, L1 = 1 m = 100 cm, T1 = 1 s, n1 = 1. In CGS (erg): M2 = 1 g, L2 = 1 cm, T2 = 1 s. Formula: n2 = n1 (M1 divided by M2)^a (L1 divided by L2)^b (T1 divided by T2)^c. [1 Mark]

• **[Step 3: Numerical Calculation]** n2 = 1 x (1000 divided by 1)^1 x (100 divided by 1)^2 x (1 divided by 1)$^{-2}$ = 10^3 × 10^4 × 1 = 10^7. [1 Mark]

• **Key Exam Result:** 1 Joule = 10^7 ergs

6. The velocity v of a particle depends on time t as v = A + B t + C divided by (D + t). Find the dimensional formulas of constants A, B, C, and D. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Dimensions of D]** In the denominator (D + t), only quantities of identical dimensions can be added: [D] = [t] = [T^1]. [0.5 Mark]

• **[Step 2: Dimensions of A and B]** By homogeneity: [A] = [v] = [L T^{-1}]. [B t] = [v] ⇒ [B] = [v] divided by [t] = [L T^{-1}] divided by [T] = [L T^{-2}]. [1.5 Marks]

• **[Step 3: Dimensions of C]** [C divided by (D + t)] = [v] ⇒ [C] divided by [T] = [L T^{-1}] ⇒ [C] = [L T^{-1}][T] = [L^1]. [1 Mark]

• **Key Exam Result:** [A] = L T^{-1}, [B] = L T^{-2}, [C] = L, [D] = T